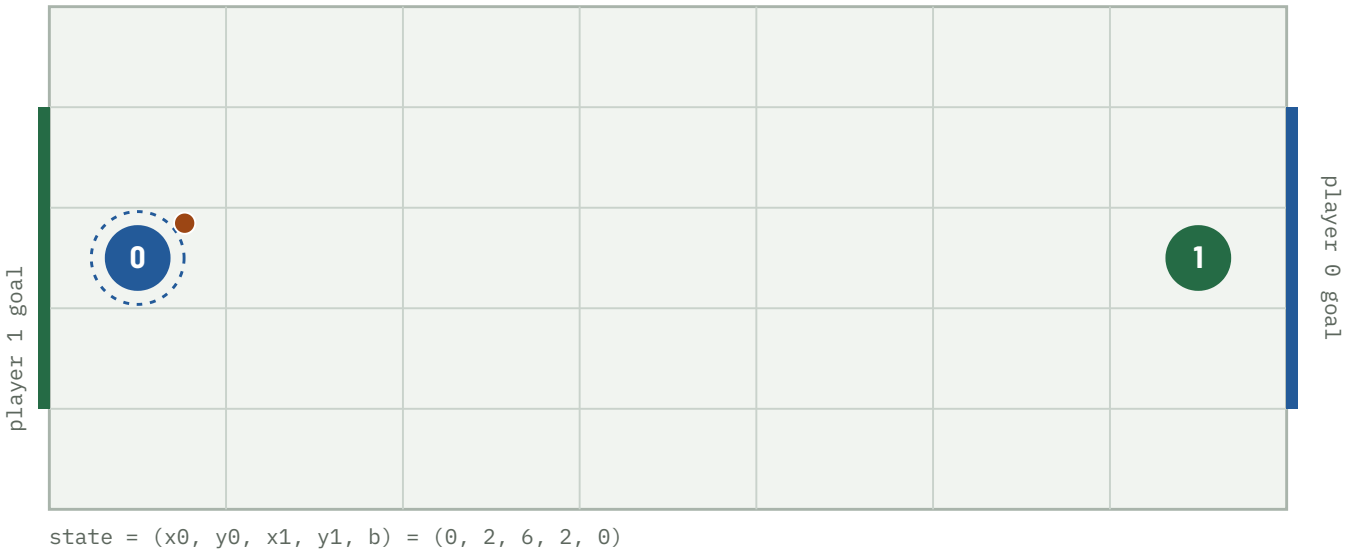
When Does Soccer Need Mixed Strategies?

A pure-first Nash-Q approach: where a two-player soccer Markov game does — and does not — need mixing, and how much LP work that saves.

FOUR QUESTIONS

RQ1 Existence — under what transition structures does a pure equilibrium exist at every state? **RQ2 Efficiency** — how much work does checking for a pure saddle first eliminate? **RQ3 Mechanism** — which spatial configurations force mixing? **RQ4 Robustness** — how do discounting and tolerance affect the split?

repo · soccer-markov-nash | 150+ tests | tables from experiments/*.csv | scope in docs/assumptions.md



The kickoff. 7×5 grid, 2 380 non-terminal states. Player 0 (blue) carries the ball and attacks the right edge; player 1 (green) the left. Actions U D L R are chosen simultaneously; reward is +1 / −1 / 0, zero-sum.

ABSTRACT

1 **For the implemented deterministic A10 model, every one of the 2380 converged stage games has a pure saddle.** Playing a pure saddle action everywhere is then a stationary pure-strategy equilibrium, found in 9 sweeps with no linear program.

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2 **A coin-flip tie-break does not change this;** Littman's random *move order* does — but only when the goal mouth is more than one cell wide, and even then on 3.95% of states (this configuration).

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3 **Those mixed states are geometrically localised:** a decision tree predicts them from the state with precision 0.96, and 68 of 94 have a 2×2 matching-pennies support.

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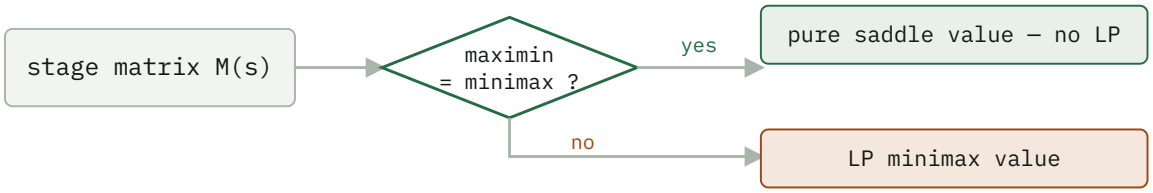
4 **The pure-first hybrid solver** matches the all-LP value function to 4e−16 while calling the LP on only 3.95% of states — ~25× fewer per sweep; mirror symmetry halves the work again.

Method

The game is zero-sum, so at every state the stage game is a payoff matrix $M(s)[a_0, a_1] = E[R + \gamma V(s')]$ and the Shapley (1953) fixed point is the minimax value:

$$V(s) = \mathrm{val}(M(s)) = \max_p \min_{k^3} (p^\top M)[a_1] = \min_p \max_{a0} (Mq)[a_0]$$

The $Q = R + \beta\,\mathrm{Nash}(Q')$ update is this operator. Three stage backups: **pure** (the maximin security value — a fast lower bound, not a Nash value when it is below the minimax), **mixed** (the LP minimax value, via `scipy` HiGHS), and **hybrid** (the pure saddle value where `maximin = minimax`, the LP elsewhere — exact everywhere). Support enumeration (`support_enum.py`) finds *all* equilibria, including of general-sum games.



The pure-first hybrid. On the deterministic game the "yes" branch is taken at every one of the 2 380 states, so the LP is never called.

RQ1 — Existence

MOVE ORDER	STOCHASTIC	CONVERGED STAGE GAMES WITH NO PURE SADDLE	STATIONARY PURE EQ?
deterministic (exact A10)	no	0 / 2380	yes (Claim B)
coinflip tie-break	yes	0 / 2380	yes (Claim B)
random move order	yes	94 / 2380 (this config)	no

IT IS THE MOVE ORDER, NOT THE RANDOMNESS

The deterministic and coin-flip value functions are *identical* (`max |V_det − V_coin| = 0` at every γ): optimal play never enters a losing contest, so the coin is never flipped. Only Littman's random move order — where a ball-steal depends on who resolves first *and* on both players' targets — creates matching-pennies subgames.

RQ2 — Efficiency

Two numbers, kept apart: the **LP-call rate** is a property of the game and reproduces exactly; **wall clock** is a median of 5 repeats and depends on the machine and LP backend (`scripts/benchmark.py`).

RANDOM MOVE ORDER, Γ = 0.9 — THE CASE WHERE THE LP IS ACTUALLY NEEDED

SOLVER	SWEEPS	LP CALLS	STATES NEEDING LP	MEDIAN WALL CLOCK
pure (lower bound)	18	0	—	0.3 s
hybrid exact	108	9 672	3.95%	13.2 s
all-LP exact	108	~3.6 M	100%	~8 min

The hybrid solves an LP on only the **3.95%** of states lacking a pure saddle — a ~25× per-sweep reduction — and matches the all-LP value to 4e−16. The wall-clock ratio (~13 s vs ~8 min) is larger but less portable: it also reflects matrix construction and the LP backend. **Mirror symmetry** (`run_symmetric`, deterministic dynamics only): every state has a distinct board-flip-plus-player-swap mirror, so the 2 380 states are 1 190 pairs — reconstruct one from the other by `V(mirror(s)) = −V(s)`, cutting the deterministic solve 0.33 s → 0.23 s.

Freeze-then-iterate, and its staleness

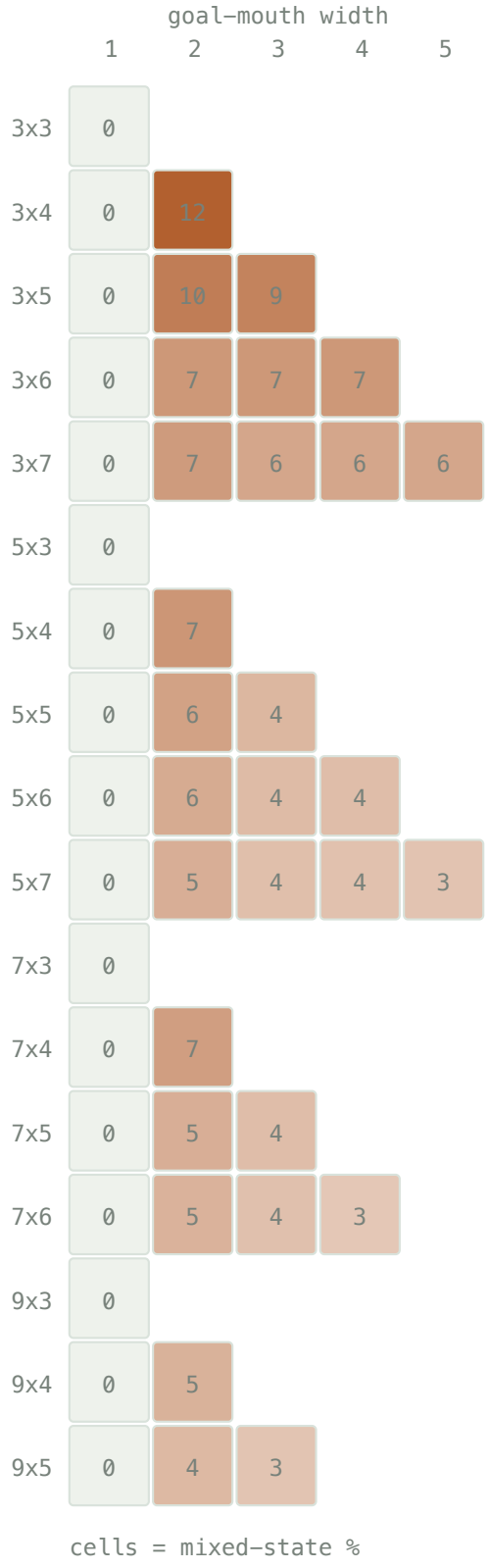
RANDOM MOVE ORDER, HYBRID SOLVER — SAME FIXED POINT, |DIFF| < 3E−9

	ROUNDS / SWEEPS	MATRIX-GAME SOLVES	WALL CLOCK
value iteration	108	9 672	13.8 s
policy iteration	21	1 879	17.2 s

5× fewer LP solves — but the frozen strategies' distance to the current Nash stays *maximal* (a full pure-strategy flip) for **16 outer rounds**, then collapses to < 1e−8. The thrashing eats the saving; on the deterministic game it is strictly worse. Value iteration with the per-sweep Nash cache wins here.

RQ3 — Mechanism

Goal-mouth width, not board size, is the gate. The phase diagram sweeps every board from 3×3 to 9×5 against goal widths 1 to `height−2` (`scripts/phase_diagram.py`, `experiments/phase_diagram.csv`).



The gate. Goal width 1 → **0** mixed states on all 16 boards: the carrier has one threat, the defender one response. Goal width ≥ 2 → mixing on every board, at 3–12% that drifts down with board area and is roughly flat in goal width beyond 2. Every state is reachable from the kickoff, so this is also the fraction over reachable states.

Geometry predicts the label (`scripts/geometry_model.py`): a depth-4 decision tree on carrier-frame features separates `mixed` from the rest with **precision 0.96, recall 0.91**. The dominant feature is `defender_can_intercept` — the defender within one move of the carrier's forward cell (`P(mixed) = 0.44` vs `0.002`).

Beyond prediction, the 94 mixed states canonicalize under the board mirror to 47 pairs and cluster into **8 geometric templates** (`scripts/templates.py`, `docs/templates.md`).

THE MIXED-STATE TEMPLATES — CARRIER-FRAME GEOMETRY, EQUILIBRIUM SUPPORT

STATES	GEOMETRY	SUPPORT	STRUCTURE
24	defender 1 ahead, carrier one row off the goal row	2×2	matching pennies
24	defender 1 ahead, same row, carrier on a goal row	2×2	matching pennies
16	defender 2 ahead, same row, carrier on a goal row	2×2	matching pennies
4	defender diagonally adjacent, carrier off the goal row	2×2	matching pennies
14	defender 1–2 ahead, same row, on a goal row	2×1	near-pure saddle
8	defender 2 ahead, same row, on a goal row	3×2	degenerate (row ties)
4	defender 2 ahead, carrier off the goal row	3×3	wider mix

The **four 2×2-support templates are all verified matching pennies** — the carrier's best reply flips between the two defender columns and vice versa — covering **68 of 94** states: carrier {climb, advance} against defender {cover a lane, hold the forward cell}. All 94 share one value across every equilibrium; 64 have a unique one.

Why the mix is unavoidable. A mixed stage game needs all three of: a stochastic (½–½) resolution order; a goal mouth wider than one cell, so the carrier has two scoring lanes; and a defender close enough to contest the forward cell but unable to cover both lanes in one move. The carrier climbs or drops, the defender guesses which — matching pennies. Drop any one clause — a single goal cell, deterministic resolution, or the coin-flip tie-break — and every stage game has a pure saddle. The full argument and one stage matrix per template are in `docs/templates.md`. For the single goal cell the theorem is partly proved (`docs/proof.md`): the defender has a closed-form optimal strategy — guard the one cell — and every stage game is dominance-solvable, machine-checked for all boards up to 11×5.

RQ4 — Numerical robustness

The value is three numbers

`value_bracket` returns what the row player guarantees ($\min_k (pM)_k$), gets ($p^\top Mq$), and can reach ($\max_i (Mq)_i$). The true value is bracketed, so the width certifies the LP error.

FINDING

With `scipy`'s HiGHS the three agree to `1.1e-16` per stage game, `8.7e-10` accumulated. The concern is real for older vertex/simplex solvers — a non-issue here.

Discounting and tolerance

	DETERMINISTIC	RANDOM (7×5)
mixed states, γ = 0.5 → 0.9 → 0.995	0 → 0 → 0	122 → 94 → 102
classification split, rel_tol 1e-12 → 1e-3	stable	604 / 1682 / 94 (stable)
classification, rel_tol 1e-2 / 1e-1	—	80 / 0 mixed
fixed 0.1-rounding flips a saddle	0	62

Discounting mildly shifts which states mix; the classification is otherwise robust across nine orders of magnitude of tolerance, breaking only as the tolerance approaches the real `minimax − maximin` gaps. Fixed 0.1-rounding — proposed to force indifference — misclassifies 62 states, exactly the small-entry mixed region.

Limitations

- The A10 page redacts the ID-specific start position and goal rows; this repo uses the standard Littman geometry (configurable). Qualitative results hold across 40+ board configurations; the exact `3.9%` is the 7×5 / 3-cell-goal case.
- Several collision sub-cases (carrier vs. stationary opponent, bump = steal, swap possession) are *interpreted* from the two stated A10 rules, not quoted. If course staff intended otherwise, Claim A must be re-measured. See `docs/assumptions.md`.
- **Claim A** (every converged stage game has a pure saddle) and **Claim B** (a stationary pure equilibrium exists) are established by enumeration. **Claim C** (the theoretical game necessarily has one, over all settings) is *not* — that needs a proof.
- `random` and `coinflip` are different game definitions, isolating the collision rule and the tie-break respectively — not the A10 game.

Secondary validation — the policy plays correctly

- Duality gap $V0\_br(s_0) + V1\_br(s_0) < 1e{-}9$ for every move order — no opponent beats the game value.
- Nash vs. Nash reproduces the value: forced draws in the deterministic and coin-flip games; `+0.157` empirical discounted return (vs. `+0.150` computed) in the random game.
- A best response to one assumed opponent is fragile: the Part 2 policy has exploitability `0.43` — worse than random — which is the A10 competition's warning about non-equilibrium submissions.

Beyond zero-sum, and open questions

- **General-sum — extension point.** `markov_game.py` solves 2-player general-sum Markov games by enumerating *all* stage equilibria (`support_enum.py`) and selecting one — the notes' "largest sum of values", a no-op for zero-sum but decisive for Battle of the Sexes, where it avoids the mixed equilibrium whose value is below either pure one. Pure-first throughout. The soccer game is zero-sum, so this does not touch the main result.
- **Function approximation — partial.** A network fit to the stage matrices (`nash_dqn.py`, 5→96→96→16 with biases, 600 epochs) trails the exact solver, over **5 seeds** (`experiments/nash_dqn_seeds.csv`):

	EXACT HYBRID NASH-Q	NEURAL NASH-Q (MEAN ± SD)
max |V − V_exact|	0	0.55 ± 0.02
mean |V − V_exact|	0	0.13 ± 0.00
action agreement	100%	43% ± 2%
pure/mixed classification	100%	67% ± 2%
exploitability	<1e−9	0.43 ± 0.04
runtime	9 sweeps, 0.3 s	600 epochs, 32 ± 1 s

The network fits the ~1 000 zero-value states easily (small *mean* error) but misses the γ^k bands and contested regions: worst-state error `0.55`, policy about as exploitable as random play, and it mis-calls *whether* a state needs mixing a third of the time. Keep the exact solver as ground truth.

- **A proof (partly done).** The single-cell pure-saddle theorem now has the defender's optimal strategy in closed form and a dominance-solvability certificate for every finite board (`docs/proof.md`); the carrier's half, board-size-free, is open.

`soccer_nash` — `game.py` · `matrix_games.py` · `support_enum.py` · `nash_q.py` · `markov_game.py` · `numerics.py` · `dominance.py` · `geometry.py` · `symmetry.py` · `tree.py` · `templates.py` · `onecell.py` · `reachability.py` · `best_response.py` · `exploit.py` · `mlp.py` · `nash_dqn.py` · `opponents.py` · `shaping.py` · `simulate.py` · `experiment.py` · `render.py`
`scripts` — `experiments.py` · `analyze.py` · `numerics.py` · `equilibria.py` · `geometry_model.py` · `templates.py` · `phase_diagram.py` · `onecell_proof.py` · `benchmark.py` · `policy_iteration.py` · `selfplay.py` · `nash_dqn.py` · `render.py` · `a10_part1.py` · `a10_part2.py` · `a10_competition.py`
`experiments` — `baseline.csv` · `gamma_sweep.csv` · `tolerance_sweep.csv` · `board_sweep.csv` ·

goal_mouth_sweep.csv · one_cell_goal.csv · phase_diagram.csv · nash_dqn_seeds.csv

env: 7×5 grid, 2 380 states, zero-sum ±1, γ = 0.9